

WHY

ODE and Simulation

How does a simulator advance the world?

$$\dot{x} = f(x, t), \quad x(t_0) = x_0$$

x state column

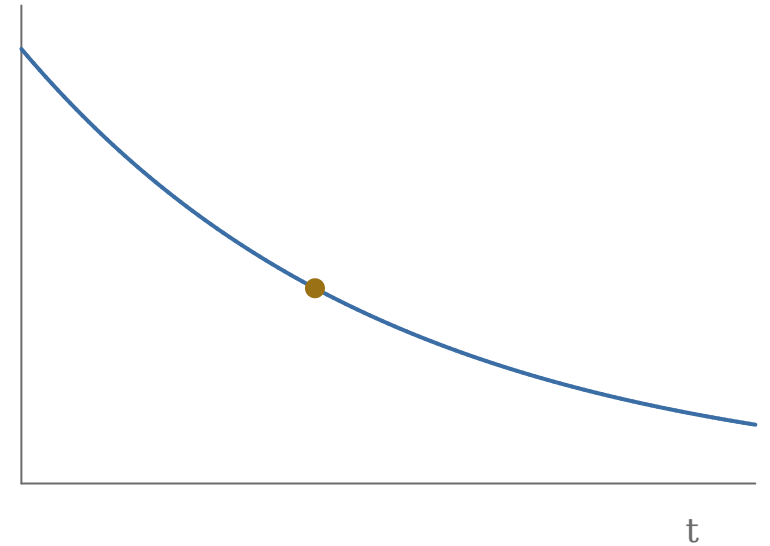
f rate of change

t time; t_0 is the start

x_0 known initial state

A model gives a velocity. A simulator must produce a future state.

x benchmark: $\dot{x} = -2x$, $x_0 = 1$



Input/state in · model out · error res · data dat

OBJECTS

ODE and Simulation

A trajectory and its stored samples

$$t_k = t_0 + kh, \quad x_k \approx x(t_k)$$

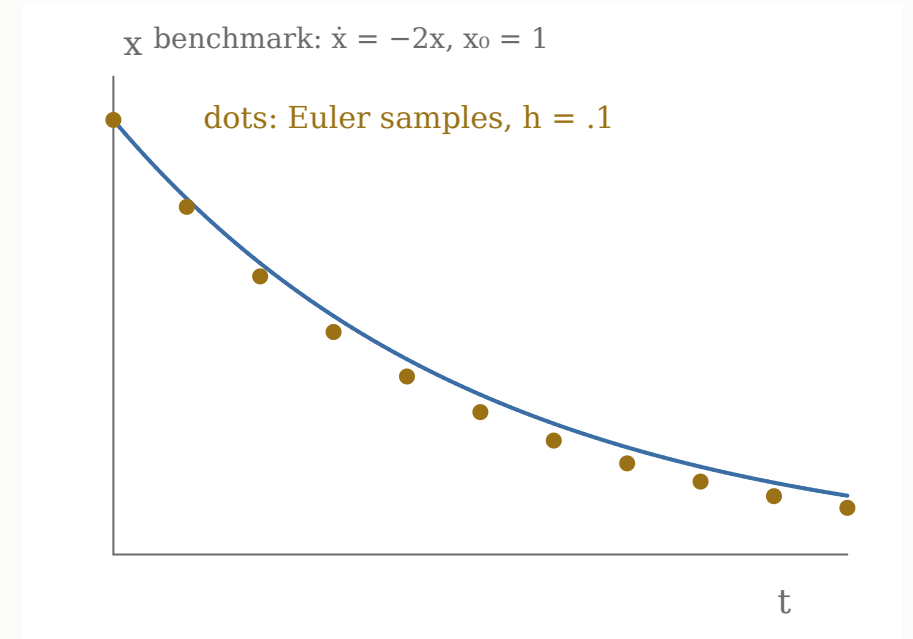
h positive time step

k integer sample index

$x(t)$ continuous exact state

x_k numerical approximation

The curve belongs to the model; the dots belong to the computation.



DEFINITION

Velocity Fields

An arrow gives a rate, not a destination

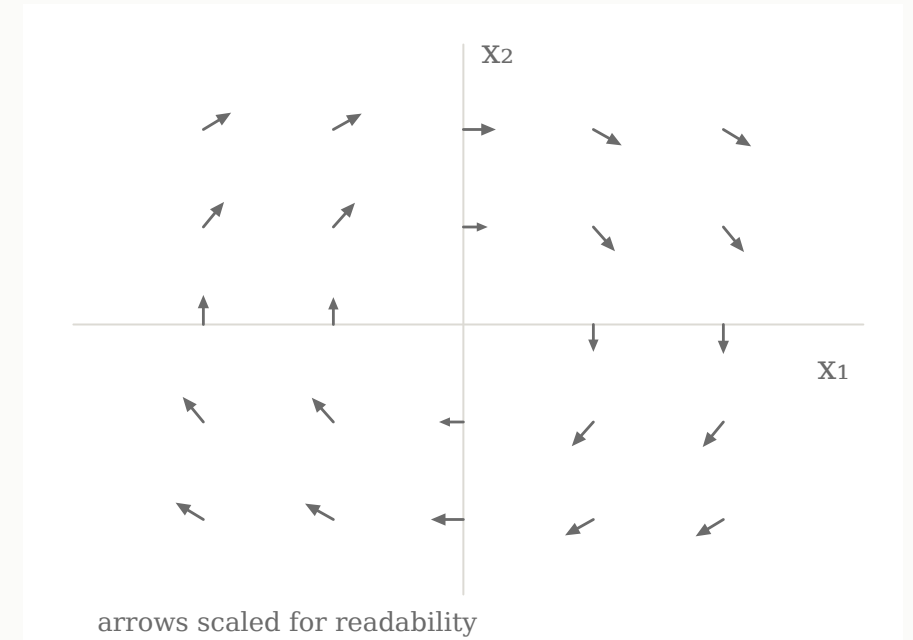
$$\dot{x} = f(x, t) \in \mathbb{R}^d$$

x d-component state column

$f(x, t)$ d-component state rate

t time held fixed in a field picture

Arrow direction and magnitude describe instantaneous motion.



ASSUMPTIONS

Velocity Fields

One field, different initial states

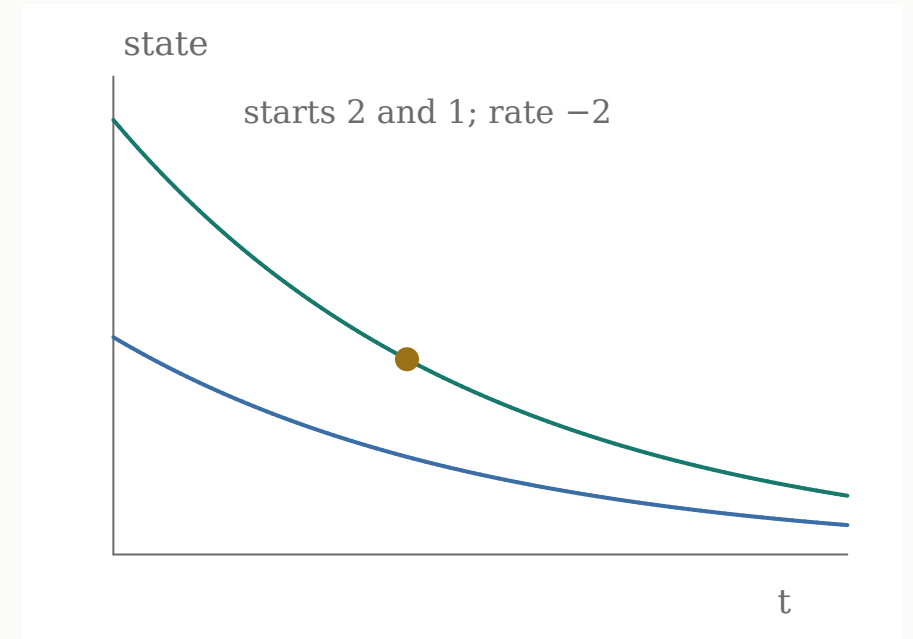
$$\dot{x} = f(x, t), \quad x(t_0) = x_0$$

x_0 chosen start

f fixed model

$x(t)$ trajectory determined by the IVP

An initial-value problem combines the rule with its starting state.



WORKED EXAMPLE

Velocity Fields

Verify an exact decay solution

$$x(t) = e^{-2t} \implies \dot{x} = -2x$$

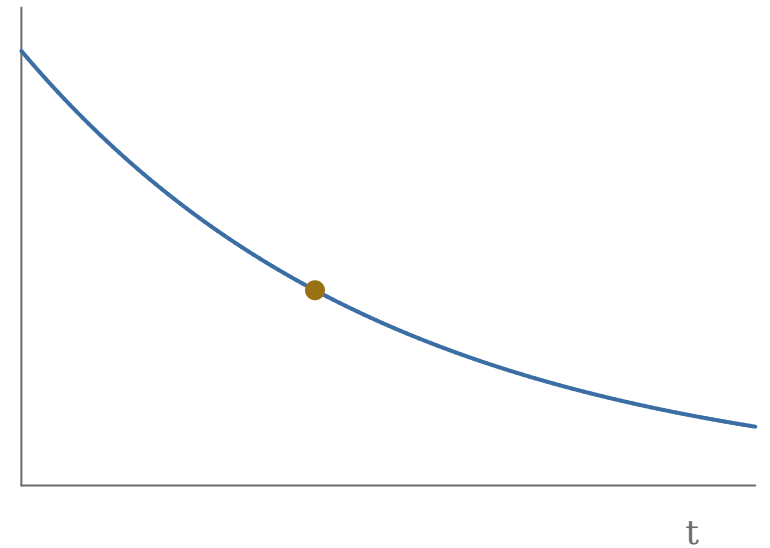
$x(0)$ one

-2 decay rate in inverse time

t elapsed time

Worked example: $x(0.1) = 0.8187307531$.

x benchmark: $\dot{x} = -2x$, $x_0 = 1$



YOUR TURN

Velocity Fields

Your turn: a changed decay rate

$$\dot{x} = -3x, \quad x(0) = 2$$

x scalar state

-3 fixed decay rate

2 initial value

Find $x(t)$, then check the differential equation and the start.

What function satisfies both conditions?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

WHY

Robot State

Position alone does not predict motion

$$x = \begin{bmatrix} q \\ v \end{bmatrix}, \quad v = \dot{q}$$

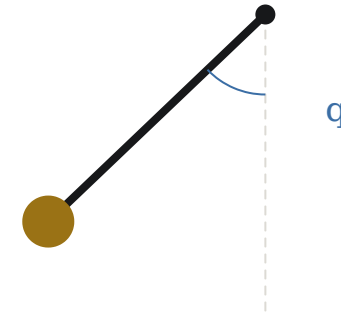
q n joint positions

v n joint velocities

x 2n-component state

The same angle can occur while a joint moves in either direction.

Schematic angle; not an integrated trajectory



downward reference

Robot State

The manipulator equation is a force balance

$$M(q)\ddot{q} + C(q, v)v + g(q) = \tau$$

M $n \times n$ inertia, positive definite

Cv velocity-dependent generalized forces

g, τ gravity and applied generalized forces

q, v positions and velocities

The model balances inertial, velocity-dependent and gravitational terms.

applied force



– velocity force



– gravity



solve with inertia

DERIVATION

Robot State

Rewrite the mechanics as a first-order ODE

$$\dot{x} = \begin{bmatrix} v \\ M(q)^{-1}[u - C(q, v)v - g(q)] \end{bmatrix}$$

x $[q;v]$, a $2n$ -column

u applied force τ

M^{-1} exists for positive-definite inertia

In code, solve $M a = u - C v - g$ rather than form the inverse.

q position



v velocity



$\dot{q} = v$



$\dot{v} = \text{acceleration}$

WORKED EXAMPLE

Robot State

A one-joint calculation

$$a = \frac{5 - 0.5(2) - 3}{2} = 0.5, \quad \dot{x} = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

M, C 2 and 0.5 at this state

g, τ 3 and 5

q, v 1 and 2

Worked example: solve acceleration, then assemble the full derivative.

q position



v velocity



$\dot{q} = v$



$\dot{v} = \text{acceleration}$

YOUR TURN

Robot State

Your turn: reduce the applied force

$$M = 2, C = 0.5, g = 3, \tau = 1, (q, v) = (1, 2)$$

M, C, g fixed at the current state

τ changed applied force

q, v unchanged state

Find both components of the state derivative.

What is the full derivative at this state?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

MODEL

Linear Dynamics

A constant matrix gives a linear field

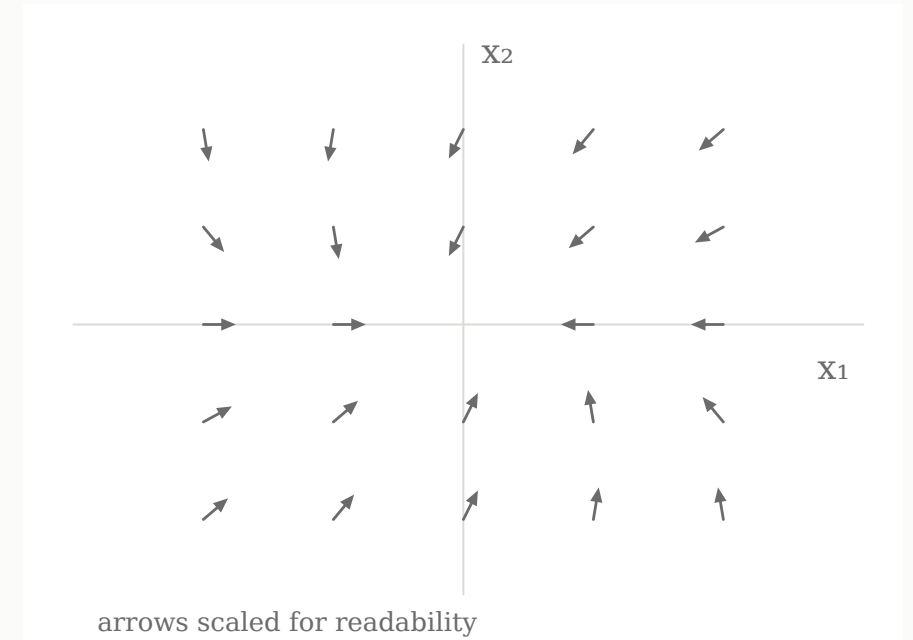
$$\dot{x} = Ax, \quad A \in \mathbb{R}^{d \times d}$$

A constant matrix

x d-component state

Ax state rate at x

No constant forcing term is hidden in this homogeneous model.



RECALL

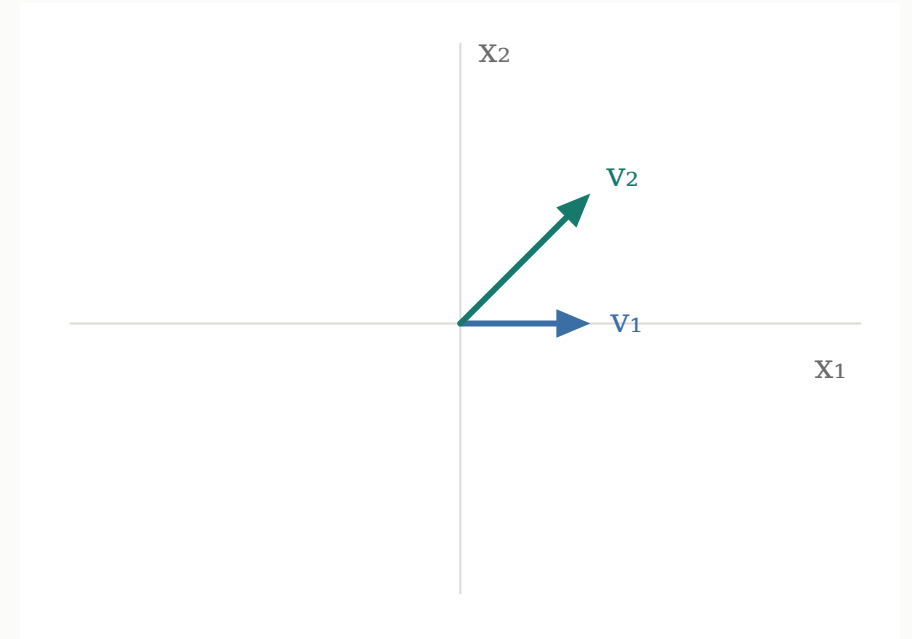
Linear Dynamics

Recall the eigenvector coordinates

$$AV = V\Lambda, \quad x = Vc$$

- V invertible eigenvector matrix
- Λ eigenvalues in matching order
- c coordinates of x in that basis

Lecture 2: a complete eigenbasis makes the modes independent.



DERIVATION

Linear Dynamics

The ODE separates into scalar modes

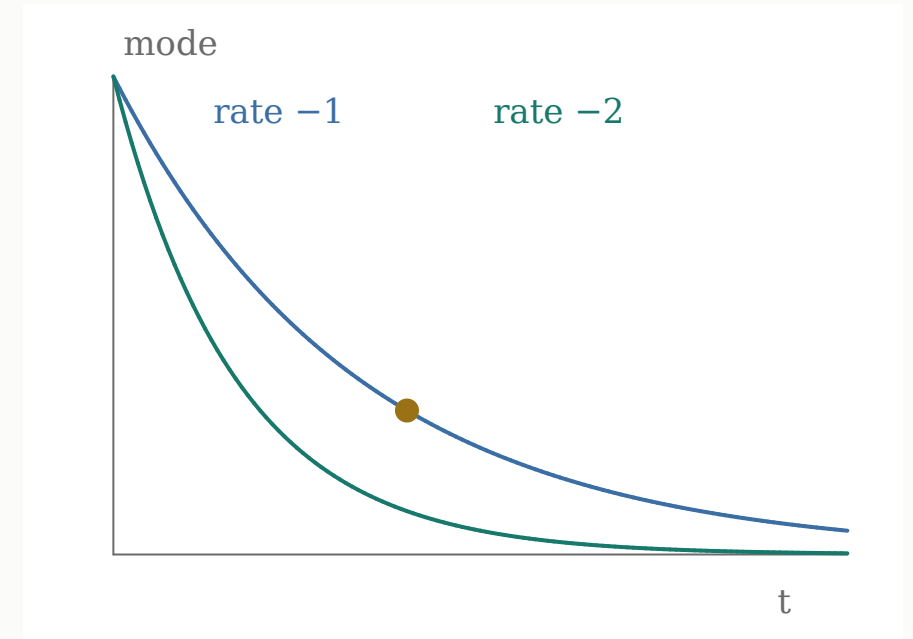
$$V\dot{c} = AVc = V\Lambda c \quad \Rightarrow \quad \dot{c} = \Lambda c$$

V constant invertible basis

c time-dependent coordinates

Λ diagonal rate matrix

Multiply on the left by V^{-1} ; each coordinate now evolves alone.



RESULT

Linear Dynamics

Each eigenvalue is a continuous growth rate

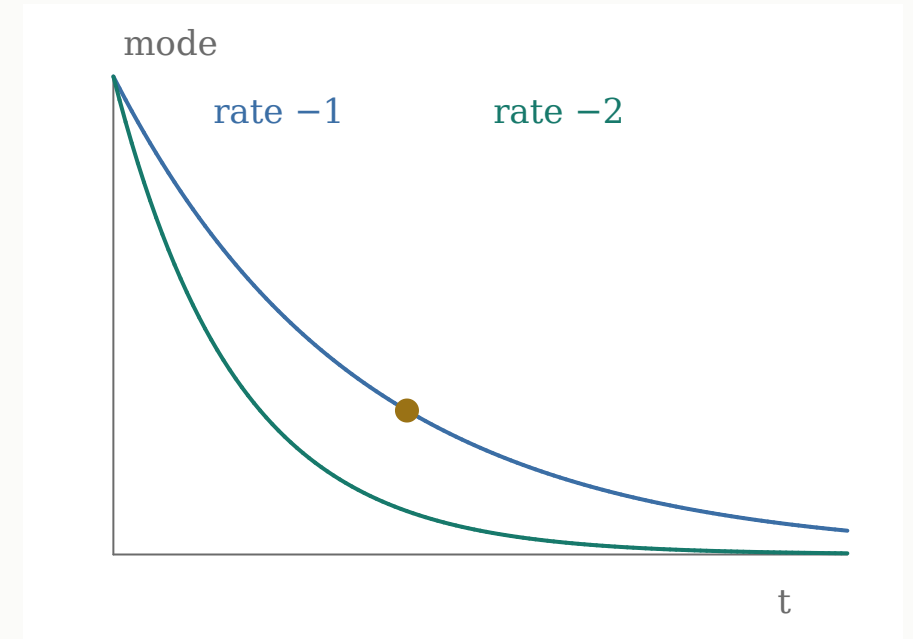
$$c_i(t) = e^{\lambda_i t} c_i(0)$$

λ_i rate of mode i

$c_i(0)$ initial amount of that mode

t time measured from zero

Negative rates decay; positive rates grow if the mode is present.



WORKED EXAMPLE

Linear Dynamics

Return from modal coordinates

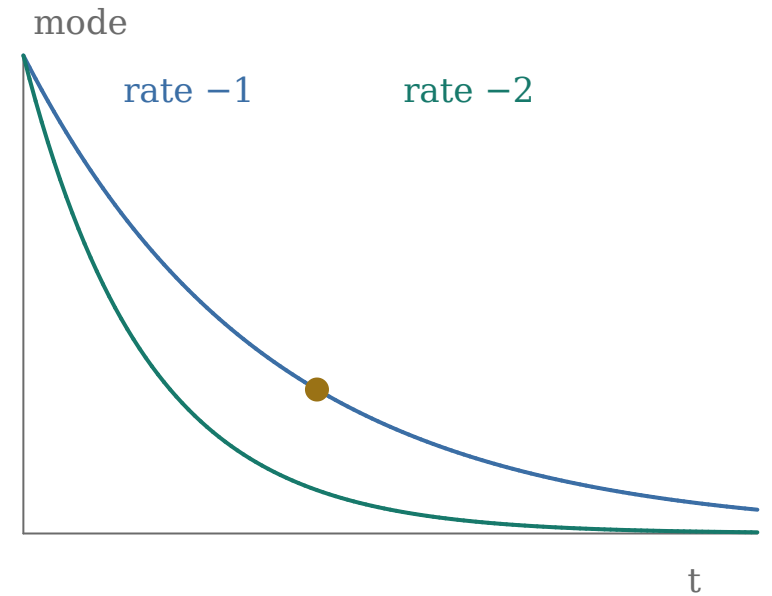
$$x(t) = V e^{\Lambda t} V^{-1} x_0 = \begin{bmatrix} e^{-t} + e^{-2t} \\ e^{-2t} \end{bmatrix}$$

V `[[1,1],[0,1]]`

Λ `diag(-1,-2)`

$V^{-1}x_0$ `(1,1)`: initial modal coordinates

$A = \begin{bmatrix} -1 & -1 \\ 0 & -2 \end{bmatrix}$, $x_0 = (2, 1)$. Check $x(0) = x_0$.



DEFINITION

Linear Dynamics

The matrix exponential always exists

$$e^{At} = \sum_{j=0}^{\infty} \frac{(At)^j}{j!}, \quad x(t) = e^{At} x_0$$

A any square constant matrix

j series index; $A^0 = I$

x_0 initial state

Diagonalization is one way to evaluate the solution, not an existence requirement.

x_0



$I + At + (At)^2/2! + \dots$



$e^{(At)} x_0$



check initial value and derivative

YOUR TURN

Linear Dynamics

A defective matrix still has a solution

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, \quad x_0 = (0, 1)^T$$

A one eigenvector direction

x_0 known initial state

t elapsed time

Test $x(t) = (t e^{-t}, e^{-t})$ by differentiation.

Does the proposed trajectory satisfy the IVP?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

WORKED EXAMPLE

Linear Dynamics

A real rotation has complex eigenvalues

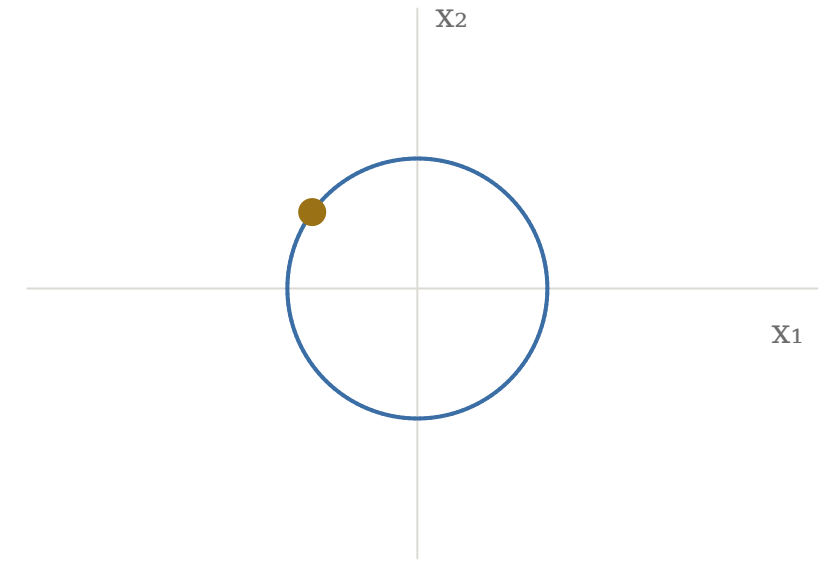
$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad x(t) = (\cos t, \sin t)^T$$

x_0 $(1,0)^T$

A constant rotation generator

λ $\pm i$ over the complex field

This trajectory stays on the unit circle.



Exact unit-circle oscillator

YOUR TURN

Linear Dynamics

Zero real parts need a closer look

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad x_0 = (0, 1)^T$$

A both eigenvalues zero

x_0 chosen initial state

$x(t)$ state to determine

Do zero eigenvalues guarantee a bounded trajectory?

What happens from the given start?

Known state → model → method



Write your rate and check its units.

The worked result appears after your attempt.

DERIVATION

Euler Integration

An exact step is an integral

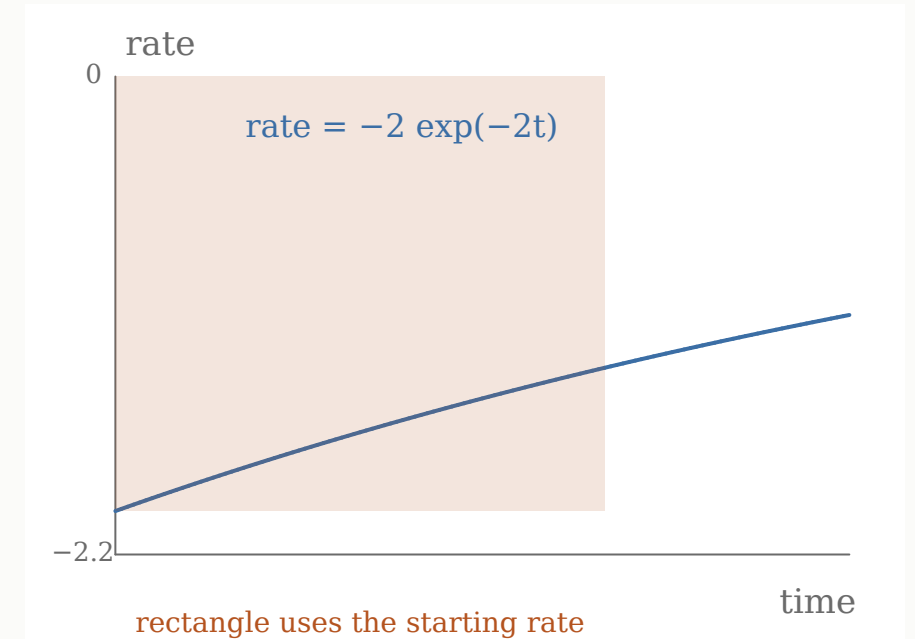
$$x(t+h) - x(t) = \int_t^{t+h} f(x(s), s) ds$$

h positive interval length

s integration variable

$f(x(s), s)$ rate along the unknown path

The fundamental theorem of calculus gives the starting relation.



RULE

Euler Integration

Euler uses the rate at the start

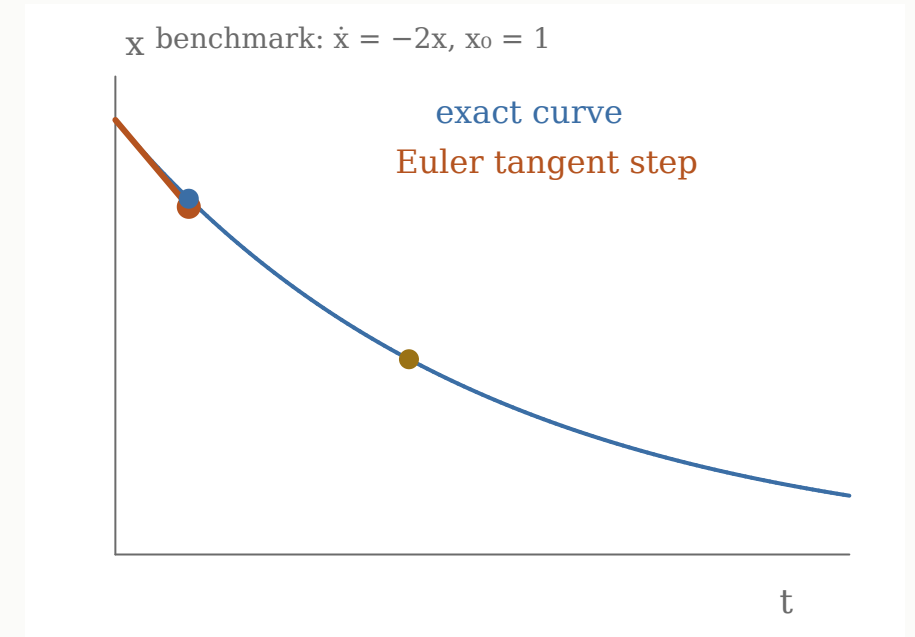
$$x_{k+1} = x_k + hf(x_k, t_k)$$

x_k current numerical state

h positive time step

$f(x_k, t_k)$ rate at the old state

Every component uses the same old state.



WORKED EXAMPLE

Euler Integration

One step against an exact answer

$$x_1 = 1 + 0.1(-2) = 0.8$$

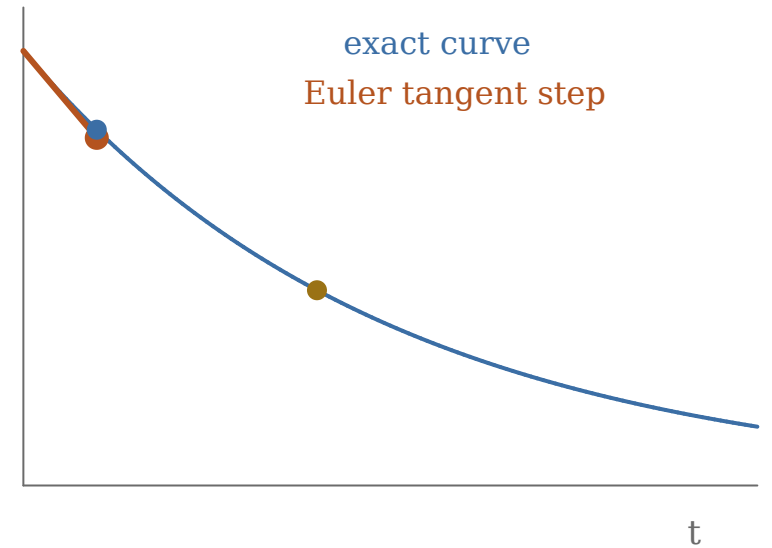
x_0 one

h 0.1

$f(x)$ $-2x$

Worked example: exact $x(0.1)=0.8187307531$; Euler is lower.

x benchmark: $\dot{x} = -2x$, $x_0 = 1$



YOUR TURN

Euler Integration

Your turn: two larger steps

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.2$$

h fixed time step

x_0 initial numerical state

x_2 approximation at $t=0.4$

Compute x_1 and x_2 . Compare x_2 with the exact state at the same time.

Which states result after one and two steps?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

Euler Integration

Local error and accumulated error differ

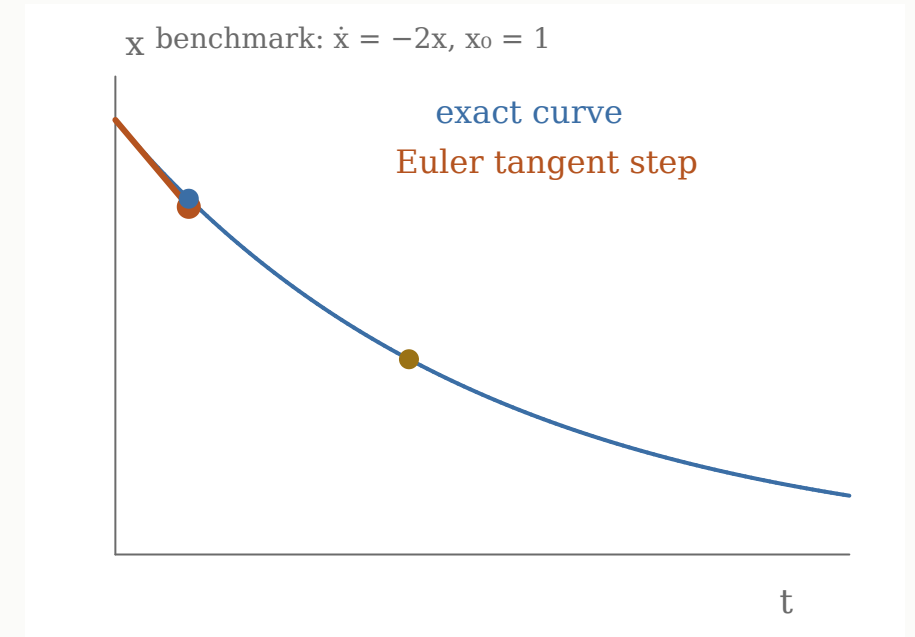
$$x(t+h) = x(t) + h\dot{x}(t) + \frac{1}{2}h^2\ddot{x}(t) + O(h^3)$$

x sufficiently smooth exact solution

h small step

$\ddot{x}$ change of the velocity along the path

Euler drops an $O(h^2)$ one-step term; fixed-time global error is $O(h)$.



Numerical Stability

A repeated step has a multiplier

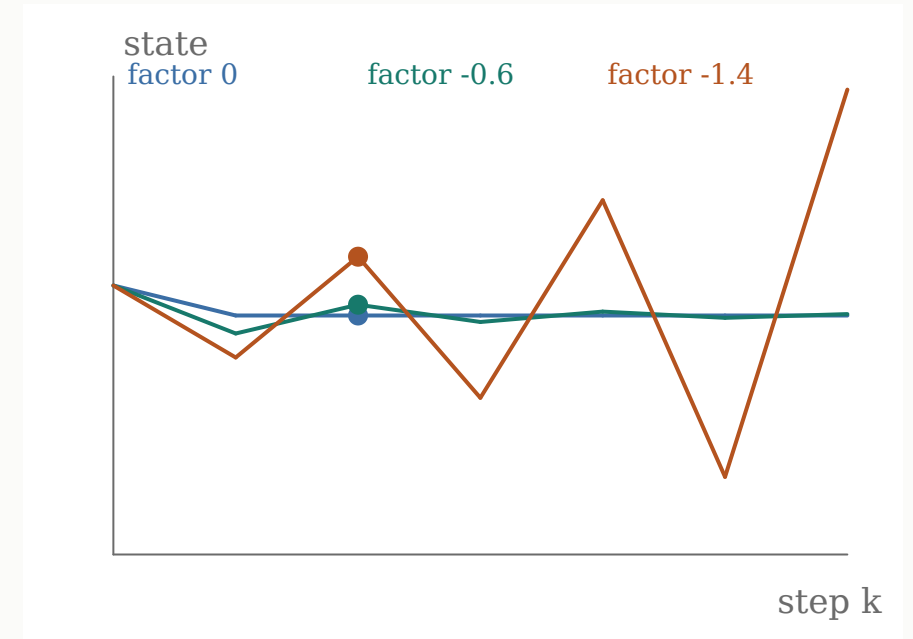
$$\dot{x} = \lambda x \quad \Rightarrow \quad x_{k+1} = (1 + h\lambda)x_k$$

λ constant scalar rate

h step size

$1+h\lambda$ numerical amplification factor

Recall Lecture 6: $A = -H$ turns this into the gradient-descent multiplier.



DERIVATION

Numerical Stability

The strict inequality decides decay

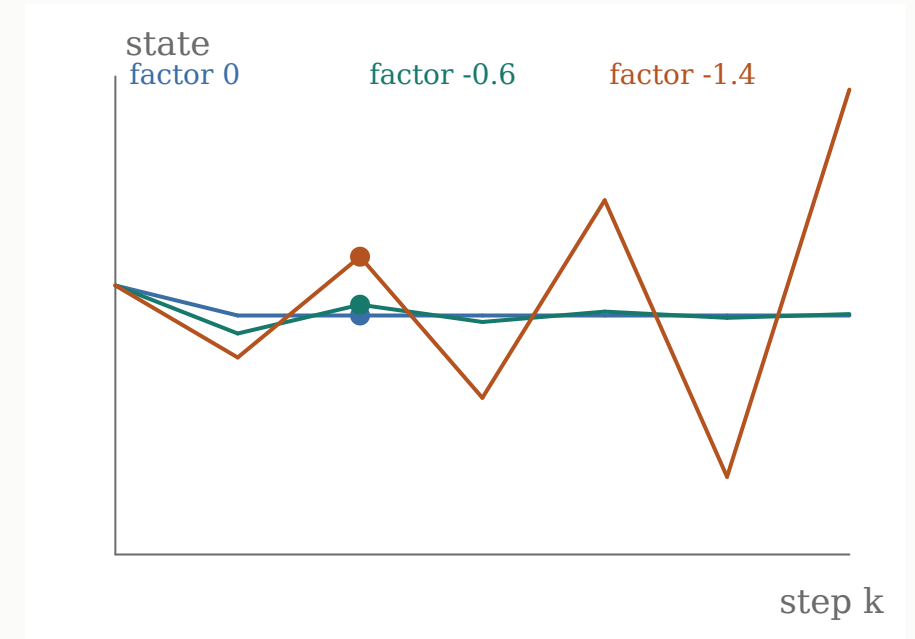
$$|1 + h\lambda| < 1 \iff 0 < h < -2/\lambda$$

λ real negative rate

h positive step size

$|1+h\lambda|$ size of each numerical multiplier

At the endpoint the factor is -1 : oscillation without decay.



YOUR TURN

Numerical Stability

Predict three step-size regimes

$$\lambda = -4, \quad h \in \{0.25, 0.4, 0.6\}$$

λ fixed decay rate

h candidate steps

x_0 any nonzero start

Classify each Euler sequence: zero, alternating decay, or growth.

What does each numerical trajectory do?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

YOUR TURN

Numerical Stability

Every mode must satisfy the bound

$$A = \text{diag}(-2, -8)$$

A constant state matrix

h one step used for both modes

x_0 arbitrary initial state

Find the step interval that makes every Euler trajectory decay.

Which condition works from every initial state?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

DIAGNOSTIC

Numerical Stability

Euler adds energy to an oscillator

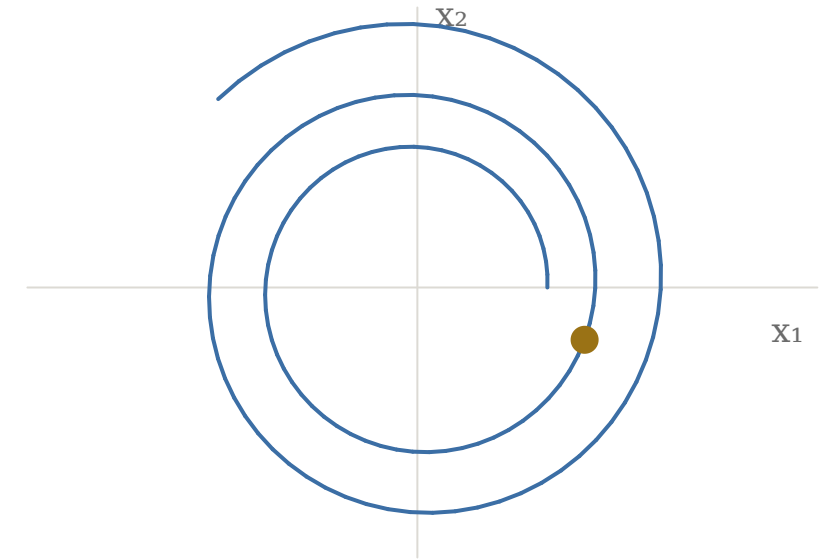
$$|1 + ih| = \sqrt{1 + h^2} > 1 \quad (h > 0)$$

$\pm i$ exact oscillator eigenvalues

h positive Euler step

$|1+ih|$ numerical amplification magnitude

A bounded exact orbit can produce an outward numerical spiral.



Euler, $h = .1$; 150 discrete updates

RECAP

Numerical Stability

What did the numerical step change?

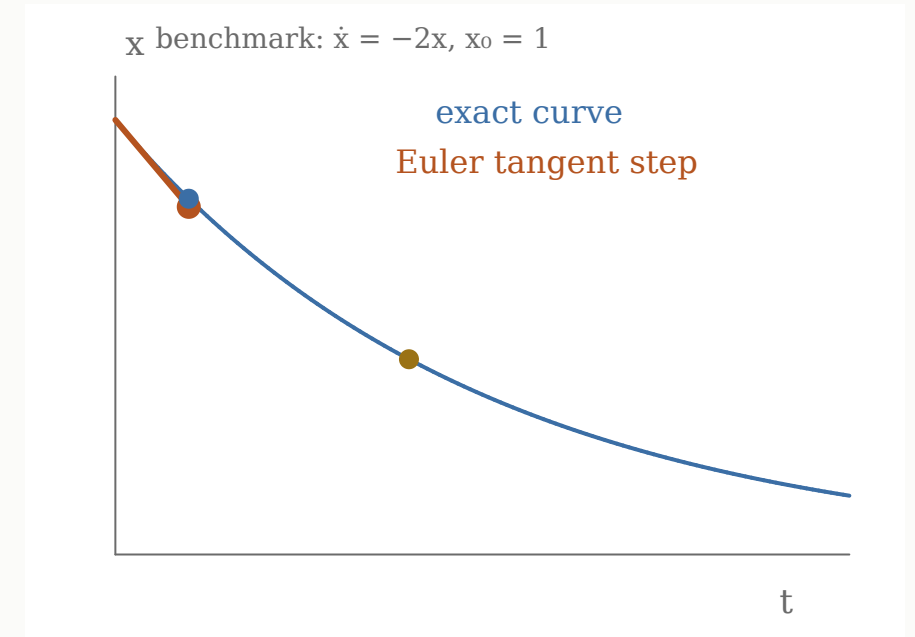
$$\text{model: } e^{\lambda h} \quad \text{Euler: } 1 + h\lambda$$

λ continuous rate

h step size

two factors exact and numerical evolution over the same interval

A correct model and a correct update formula can still give a poor simulation.



WHY

Runge-Kutta Methods

Sample the field inside the step

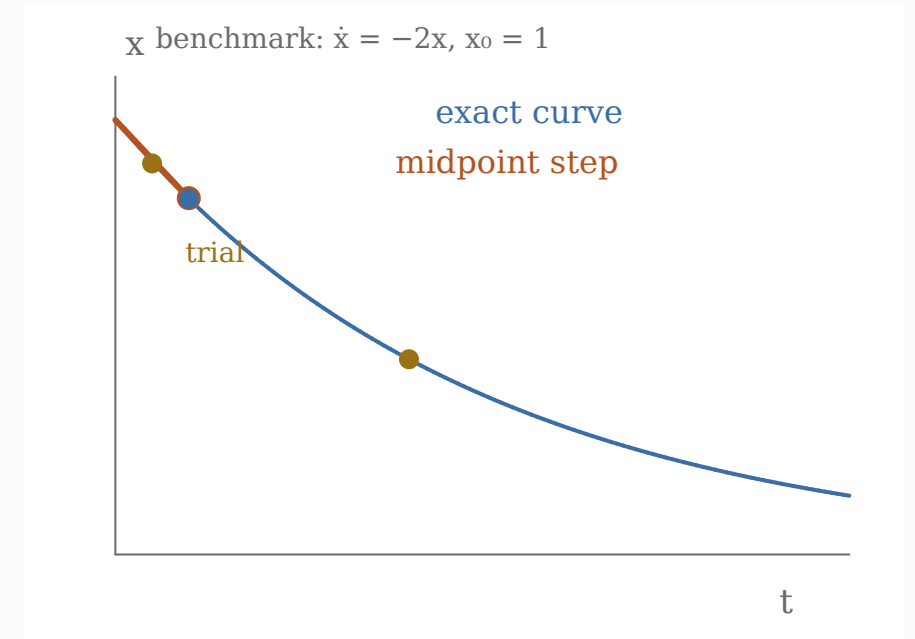
$$\Delta x = \int_{t_k}^{t_k+h} f(x(s), s) ds$$

Δx exact increment from the current exact state

f same velocity model

h one integration interval

A better slope estimate can improve the same time step.



RULE

Runge-Kutta Methods

Construct a trial midpoint

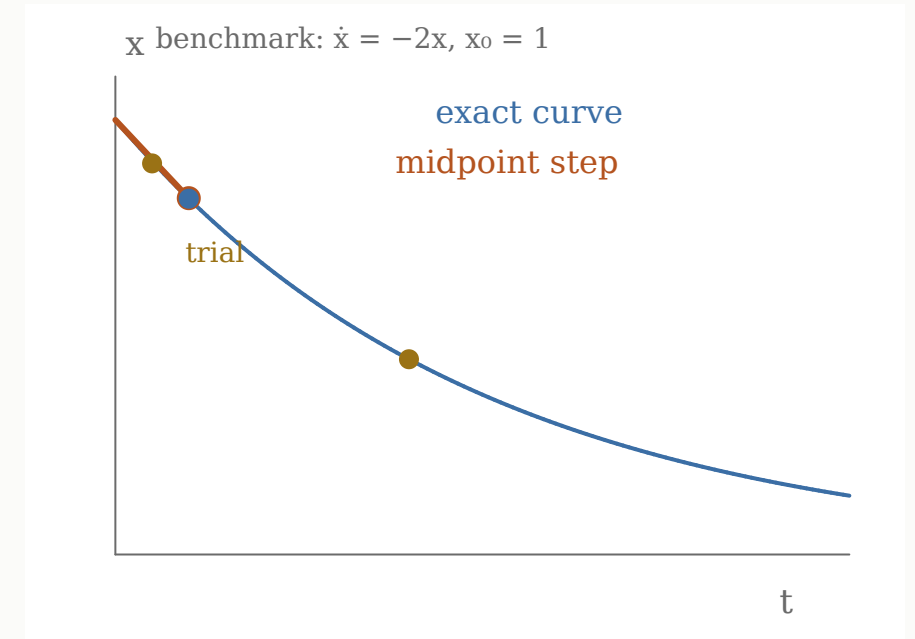
$$k_1 = f(x_k, t_k), \quad k_2 = f\left(x_k + \frac{h}{2}k_1, t_k + \frac{h}{2}\right)$$

k_1, k_2 rates, not increments

h $k_1/2$ trial half-step displacement

$t_k + h/2$ midpoint time

The first rate predicts a location; the second samples the field there.



WORKED EXAMPLE

Runge-Kutta Methods

Advance with the midpoint rate

$$x_{k+1} = x_k + hk_2$$

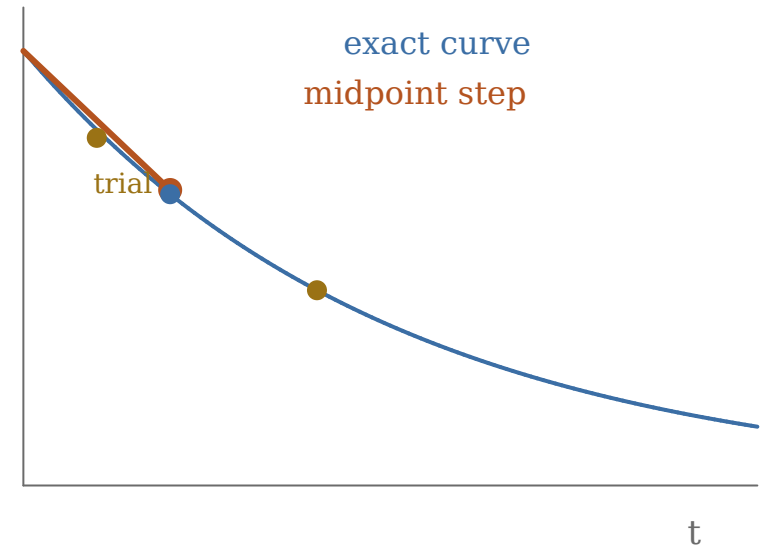
x_k original current state

k_2 rate at the trial midpoint

h full step length

$\dot{x} = -2x$, $x_0 = 1$, $h = .2$: $k_1 = -2 \rightarrow \text{trial} = .8 \rightarrow k_2 = -1.6 \rightarrow x_1 = .68$.

x benchmark: $\dot{x} = -2x$, $x_0 = 1$



YOUR TURN

Runge-Kutta Methods

Your turn: halve the midpoint step

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.1$$

k_1, k_2 two model evaluations

h full step

x_1 accepted next state

Write both stages and the accepted state.

What is the accepted midpoint result?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

RULE

Runge-Kutta Methods

Classical RK4 evaluates four trial rates

$$k_1 = f(x, t)$$

$$k_2 = f(x + hk_1/2, t + h/2)$$

$$k_3 = f(x + hk_2/2, t + h/2)$$

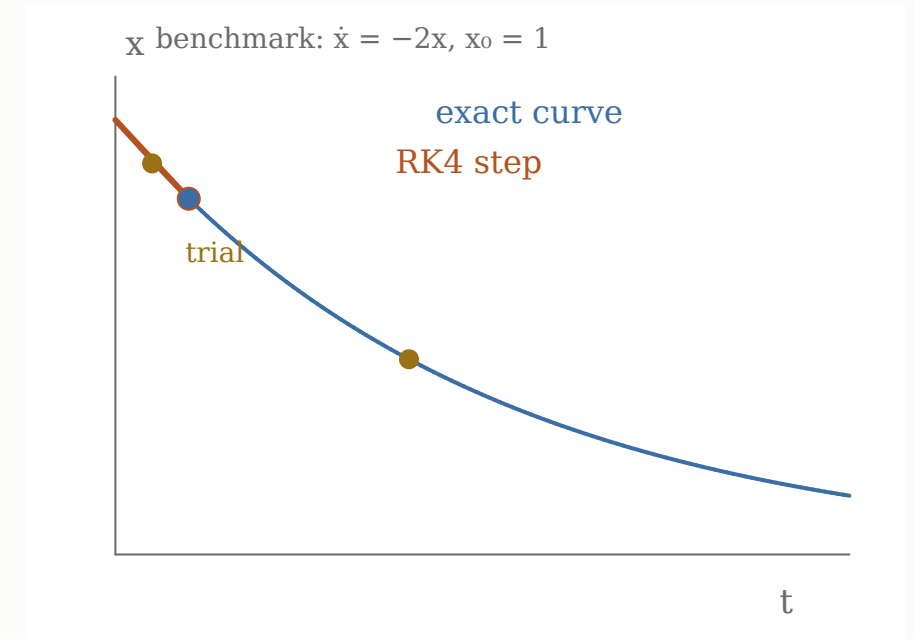
$$k_4 = f(x + hk_3, t + h)$$

x, t current state and time

h one full step

$k_1 \dots k_4$ four rates with state/time units

Each trial is anchored at the original x .



WORKED EXAMPLE

Runge-Kutta Methods

Combine the four rates

$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

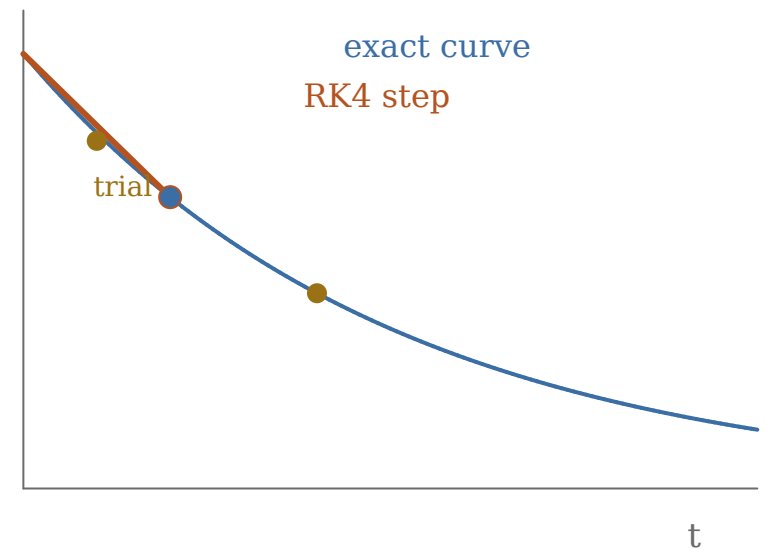
$k_1 \dots k_4$ RK4 rates

1,2,2,1 weights before division by six

h full step length

$\dot{x} = -2x$, $x_0 = 1$, $h = .2$: rates $(-2, -1.6, -1.68, -1.328)$ give $x_1 = .6704$.

x benchmark: $\dot{x} = -2x$, $x_0 = 1$



YOUR TURN

Runge-Kutta Methods

Your turn: RK4 and model-call cost

$$\dot{x} = -2x, \quad x_0 = 1, \quad h = 0.1$$

`h` full step

`k1...k4` rates to compute

`cost` number of calls to `f`

Compute one RK4 step and count model calls.

What does one step return, at what model-call cost?

Known state → model → method



Write your rate and check its units.

The worked result appears after your attempt.

COMPARISON

Runge-Kutta Methods

Accuracy improves at a computational price

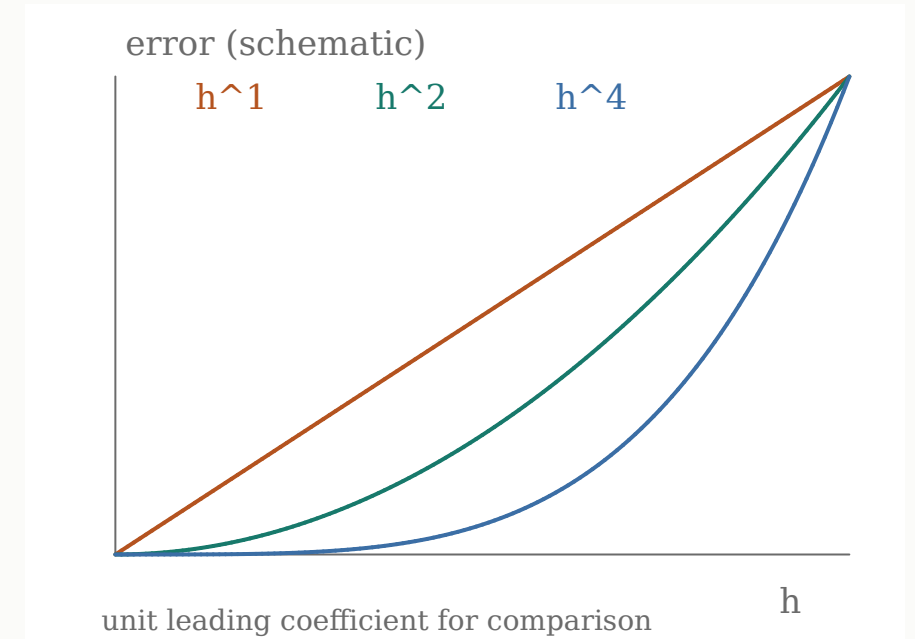
$$E_{\text{Euler}} = O(h), \quad E_{\text{midpoint}} = O(h^2), \quad E_{\text{RK4}} = O(h^4)$$

E fixed-final-time global error

h step tending to zero

assumptions sufficient smoothness and a well-posed finite interval

Halve h and check convergence; RK4 is not unconditionally stable.



MODEL

Pendulum Simulation

The restoring acceleration has a sign

$$\dot{q} = v, \quad \dot{v} = -\frac{g_0}{\ell} \sin q$$

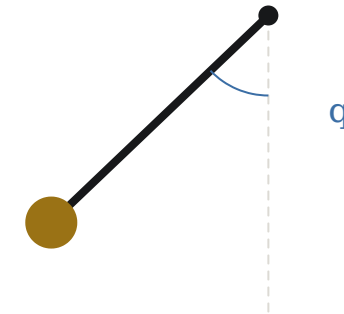
q angle from downward vertical

v angular velocity

g_0, ℓ positive gravity acceleration and length

Unforced, undamped motion is the model for this comparison.

Schematic angle; not an integrated trajectory



downward reference

DEFINITION

Pendulum Simulation

Mechanical energy is a diagnostic

$$E(q, v) = \frac{1}{2}m\ell^2v^2 + mg_0\ell(1 - \cos q)$$

first term kinetic energy

second term potential energy above the bottom

m, ℓ, g_0 fixed positive constants

The zero of potential energy is a convention; energy differences have meaning.

energy

Potential, $v = 0$

$m = \ell = g_0 = 1$; add $v^2/2$

q

DERIVATION

Pendulum Simulation

The exact dynamics conserve energy

$$\dot{E} = m\ell^2 v \dot{v} + mg_0 \ell \sin q \dot{q} = 0$$

$\dot{q}=v$ kinematic identity

$\dot{v} = -(g_0/\ell)\sin q$

$\dot{E}$ rate of mechanical energy

The two terms cancel only for the stated unforced, undamped model.

energy

Potential, $v = 0$

$m = \ell = g_0 = 1$; add $v^2/2$

q

Integrated Pendulum Problem

Choose the state and the check first

$$m = \ell = g_0 = 1, \quad x_0 = (1, 0)^T, \quad h = 0.1$$

q_0 one radian

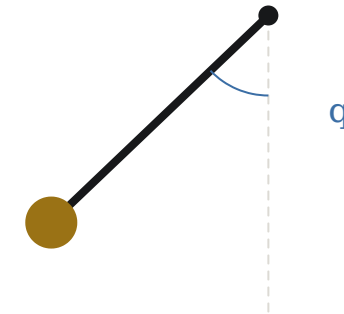
v_0 zero

h one time step

E $v^2/2 + 1 - \cos q$

Find one numerical step, then test its energy.

Schematic angle; not an integrated trajectory



downward reference

WORKED EXAMPLE

Integrated Pendulum Problem

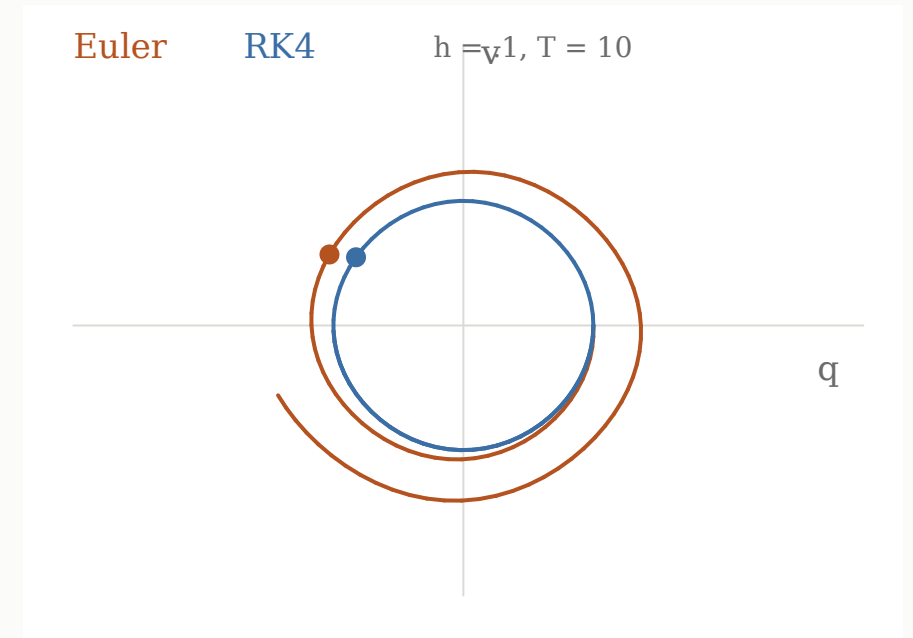
Compare the first accepted states

method	q_1	v_1
Euler	1	-0.08414710
midpoint	0.99579265	-0.08414710
RK4	0.99579454	-0.08407114

q_1, v_1 accepted state after $h=.1$

all methods same model and initial state

Euler leaves q unchanged on its first step because $v_0=0$.



EXPERIMENT

Integrated Pendulum Problem

Repeat to the same final time

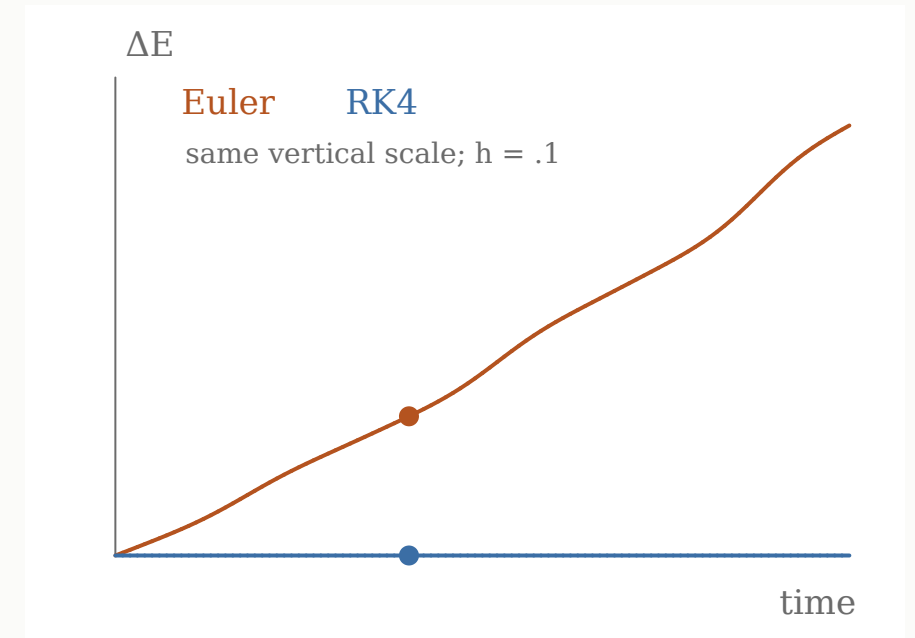
h	ΔE_{Euler}	ΔE_{RK4}
.2	1.324213	-1.46×10^{-5}
.1	.539746	-4.61×10^{-7}
.05	.232375	-1.44×10^{-8}

ΔE $E(x_N) - E(x_0)$

T 10, fixed for every run

N T/h steps

Same initial state, same dynamics, same endpoint; only method and h change.



INTERPRETATION

Integrated Pendulum Problem

Energy error does not measure phase error

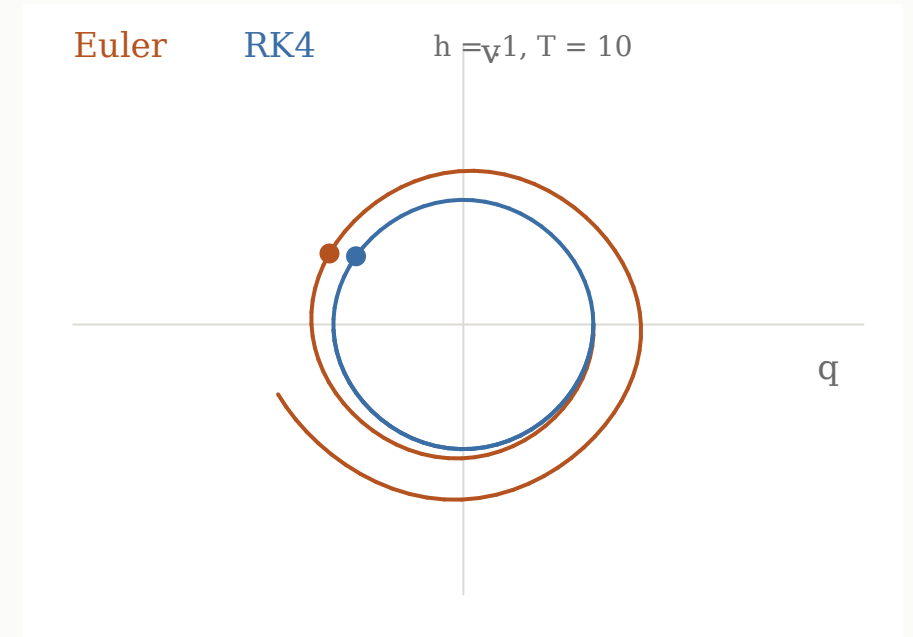
$$E(x_k) \approx E(x_0) \not\Rightarrow x_k \approx x(t_k)$$

E one scalar diagnostic

x_k two-component numerical state

$x(t_k)$ exact state at the same time

Compare angle, velocity and energy; then halve the step.



YOUR TURN

Independent Transfer

A different angle, the same method choice

$$q_0 = 0.5, \quad v_0 = 0, \quad h = 0.1, \quad m = \ell = g_0 = 1$$

q_0 new angle in radians

v_0 zero angular velocity

h Euler step

Compute one Euler step and its energy change.

Does the first Euler step preserve energy?

Known state $\rightarrow$ model $\rightarrow$ method



Write your rate and check its units.

The worked result appears after your attempt.

YOUR TURN

Simulation Checklist

Cold-write a complete simulation step

$$\dot{x} = f(x, t), \quad x(t_0) = x_0$$

x complete state

f model rate

h chosen integration step

Name the state, update, stability question and diagnostic.

What four checks make the result interpretable?

Known state → model → method



Write your rate and check its units.

The worked result appears after your attempt.

CONNECTION

Simulation Checklist

A learned field still needs an integrator

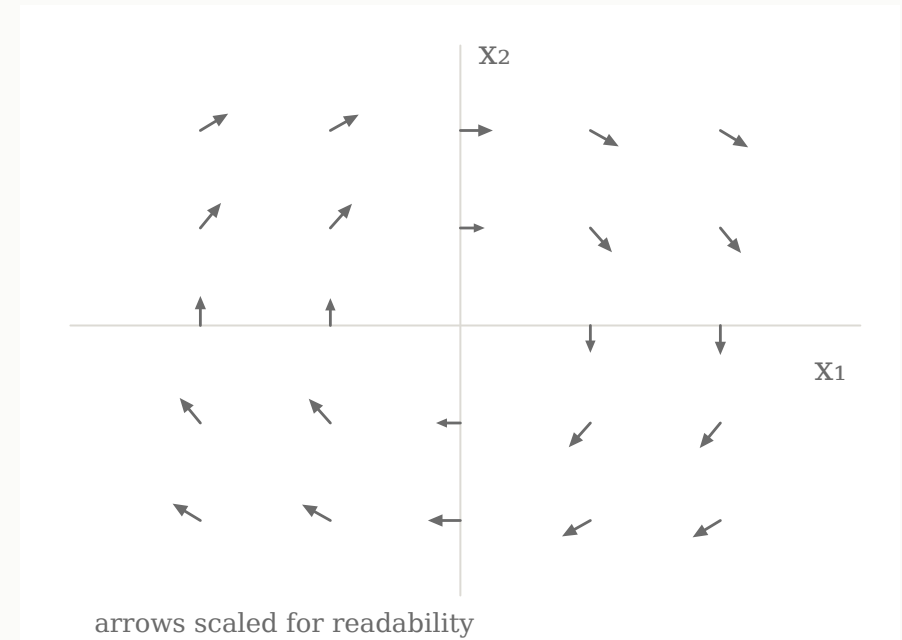
$$\dot{x} = v_{\theta}(x, t)$$

v_{θ} a field represented by a learned model

x generated state

t continuous evolution coordinate

Later: flow matching supplies a field; Euler or RK advances it.



References

Sources for models and exact dynamics

1. [Tedrake, R. Underactuated Robotics, ch.2. MIT course notes, 2024.](#)
2. [Lynch, K. M. & Park, F. C. Modern Robotics, ch.8. Cambridge, 2017.](#)
3. [Strang, G. Differential Equations and \$\exp\(At\)\$. MIT 18.06 Lecture 23, 2010.](#)

References

Sources for integration and stability

1. [Kværnø, A. Numerical Solution of ODEs, part 2. NTNU, 2022.](#)
2. [Driscoll, T. & Braun, R. Fundamentals of Numerical Computation, §6.4. Online textbook.](#)